

Technical Decision Memo

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TO	Investment Committee — Series B lead investor	DATE	September 2026
FROM	DY Research	ENGAGEMENT	DYR-DD-2026-009 · Due Diligence
RE	Meridane Robotics — the MR-7 joint controller	READING TIME	Four minutes

THE TECHNICAL ANSWER

The technology is real. Two of its claims are not.

The architecture and the engineering hold up. Two published claims do not hold as built; both are fixable in software, and neither changes what the technology is. One changes when it can ship under a safety certificate.

886	2,970	488	0
µs worst-case servo latency claimed: under 500	µs fault to stop claimed: under 2,000	µs after two contained software changes	findings that change the technical thesis

What holds

- ✓ **The architecture.** Fixed-priority Rust firmware with no allocator on any control path. A sound base for a certifiable product.
- ✓ **The engineering.** Every push is built and tested; 31 hardware-in-the-loop tests run nightly; builds reproduce bit for bit.
- ✓ **The licences.** All 114 dependencies are permissive. Nothing copyleft ships in the binary.

What does not

- **“Latency under 500 µs.”** 886 µs as built: a 180 µs telemetry lock blocks the servo loop. Its budget is 18 µs.
- **“Safe stop within 2 ms.”** 2,970 µs: faults are polled every 2 ms. No priority change can fix a polling period.
- **The timing margin.** After the fixes, 12 µs, measured against bench maxima rather than proven bounds.

The five risks that matter

RISK	LIKELIHOOD	IMPACT	MITIGATION
Safety certification slips. The safety concept relies on a 2 ms stop the design cannot deliver.	HIGH	HIGH	Verify: V 2.
A published claim fails in the field. The 500 μs figure is in the datasheet and in two customer contracts.	HIGH	MEDIUM	Verify: V 1.
The margin is thinner than measured. 12 μs rests on a 210 μs execution time that is a bench maximum.	MEDIUM	MEDIUM	Verify: V 3.
Key people. Two engineers wrote 81% of the control firmware.	MEDIUM	HIGH	Retention terms; a documentation milestone.
A single encoder supplier. No second source is qualified.	LOW	MEDIUM	Qualify a second source within 12 months.

Before relying on the claims, verify

V 1 Telemetry off the shared lock. The servo loop's response under 500 μs at every phasing, shown on the bench with a GPIO test.	V 2 Stop on an interrupt. Fault to gate-driver disable under 50 μs on a scope, and the safety concept revised to match.	V 3 A bound, not a maximum. The servo task's execution time with a stated method, clearing 222 μs with margin, with a stated margin.
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Effect on the technical thesis

None of the findings changes what the technology is or what it can become. Two change when it can ship under a safety certificate: on the evidence, the safety track needs six to eight more weeks than planned.

Where this memo stops

It summarises a three-week technical due diligence and the full report with its evidence appendix. It is not investment advice. The commercial and investment decision remains with the client.

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This is a sample memo. Every report we write belongs to its client and is published only with their written consent, so Meridane Robotics and every fact about it here are fictional. The analysis and the arithmetic are real, and match the sample Focused Audit figure for figure.